LAB 1

Inter-City Freight Load Matching Marketplace

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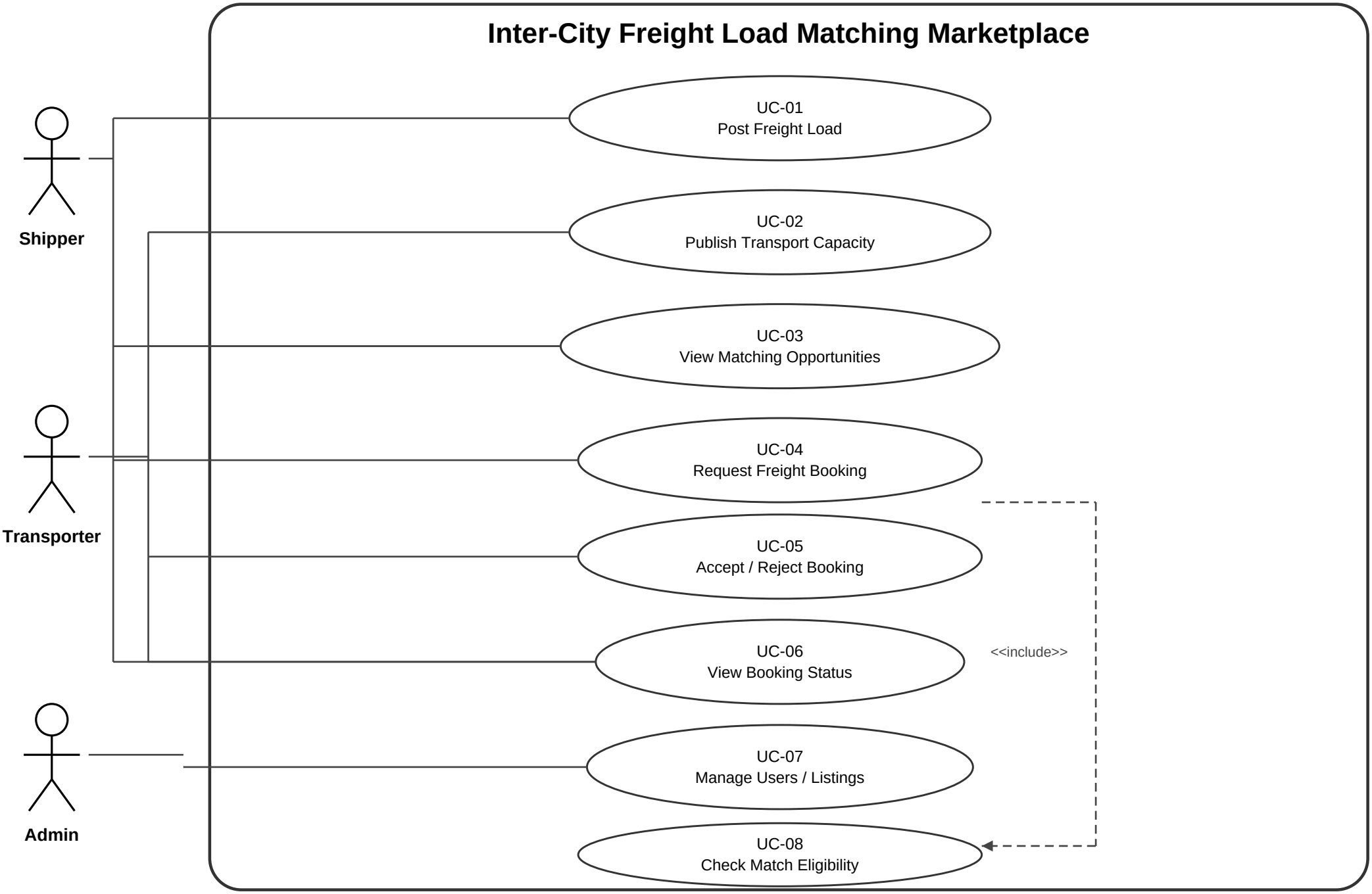
LAB 1 - REQUIREMENTS TABLE

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Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow a shipper to create and publish a freight load request containing source city, destination city, cargo type, required capacity/weight, and pickup date.	High	Given valid mandatory load details, when the shipper submits the request, the system stores it and displays the load with status "Open".	Core input needed to place freight into the marketplace.
FR-002	Functional	The system shall allow a transporter to publish an available transport capacity listing containing source city, destination city, vehicle/capacity details, and availability date.	High	Given valid mandatory capacity details, when the transporter publishes the listing, the system stores it and displays the listing with status "Available".	Provides the transportation supply that can be matched with freight.
FR-003	Functional	The system shall identify and display matching freight loads and transport capacity using source, destination, capacity/weight, and date compatibility.	High	For a test dataset containing at least one compatible and one incompatible listing, the system displays the compatible listing(s) and excludes the incompatible listing(s) from the match results.	This is the central function of the load-matching marketplace.
FR-004	Functional	The system shall allow a shipper to submit a booking request for a selected compatible transport capacity listing.	High	When the shipper selects an available compatible listing and submits the request, the system creates one booking request with status "Pending" and records the linked load and transport listing.	Converts a match into a bookable transaction.
FR-005	Functional	The system shall allow a transporter to accept or reject a pending booking request and update the booking status accordingly.	High	For a pending request, selecting Accept changes the status to "Confirmed" and selecting Reject changes it to "Rejected"; the resulting status is visible to the shipper.	Completes the matching-to-booking workflow and provides a clear outcome.
NFR-001	Nonfunctional	The system shall return matching results within 3 seconds for at least 95% of match queries during a test run of 100 queries.	High	Run 100 match queries and record response time; at least 95 queries complete in less than 3 seconds.	Provides responsive matching for practical marketplace use.
NFR-002	Nonfunctional	The system shall prevent unauthenticated access to booking-management functions.	High	In a security test of 50 unauthenticated requests to booking-management functions, all 50 requests are denied and no booking data or actions are exposed.	Protects booking data and actions from unauthorized users.

Exactly 5 FRs and 2 NFRs; each row uses a "shall" statement and measurable pass/fail acceptance criteria.



Actors: Shipper, Transporter, Admin | 8 use cases | Mandatory reuse shown with <<include>>

LAB 1 - USE-CASE FLOW

UC-05: Request Freight Booking

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Primary Actor	Shipper
Supporting Actor	Transporter
Preconditions	A compatible transport-capacity listing is visible to the shipper and is marked Available; the shipper is authenticated.
Postcondition	The booking is Confirmed when the transporter accepts; otherwise the booking ends in Rejected or the request is cancelled when the selected listing is no longer available.

Main Success Scenario

1. Shipper opens the list of matching transport-capacity listings.
2. System displays available listings that match the load on source, destination, capacity/weight, and date compatibility.
3. Shipper selects a suitable transport-capacity listing.
4. System displays the selected listing details and current availability status.
5. Shipper submits a freight booking request.
6. System validates the request and verifies that the selected listing is still Available.
7. System records the request with status Pending and notifies the transporter.
8. Transporter reviews the request and selects Accept.
9. System changes the booking status to Confirmed and makes the confirmation visible to the shipper.
10. Use case ends successfully.

Alternate Flow

6a. Selected listing no longer available

6a1. System detects that the selected listing is no longer Available.
6a2. System does not create the booking request.
6a3. System informs the shipper that the listing is unavailable and returns the shipper to the matching-listings view.

8a. Transporter rejects request

8a1. Transporter selects Reject.
8a2. System changes the booking status to Rejected.
8a3. System informs the shipper and the use case ends without confirmation.